CANCELLATION & REVENUE LOSS ANALYSIS
Full Year 2022

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- **Stakeholder**: Rahul Kumar, Revenue Assurance Manager at Amazon India, responsible for reducing revenue leakage across operations. (Perplexity Role Play)

- **Business Problem:** The Revenue Assurance Manager at Amazon India is concerned about revenue leakage from cancelled orders despite overall business growth. He wants to understand the scale of financial loss with actual numbers, "how many orders are being cancelled", "what is the total financial impact", and an assessment of whether the Cancellation & Revenue Loss rates are at an alarming stage or within a manageable range. He also needs to identify which geographies, product categories, sizes, and shipping types require priority attention.

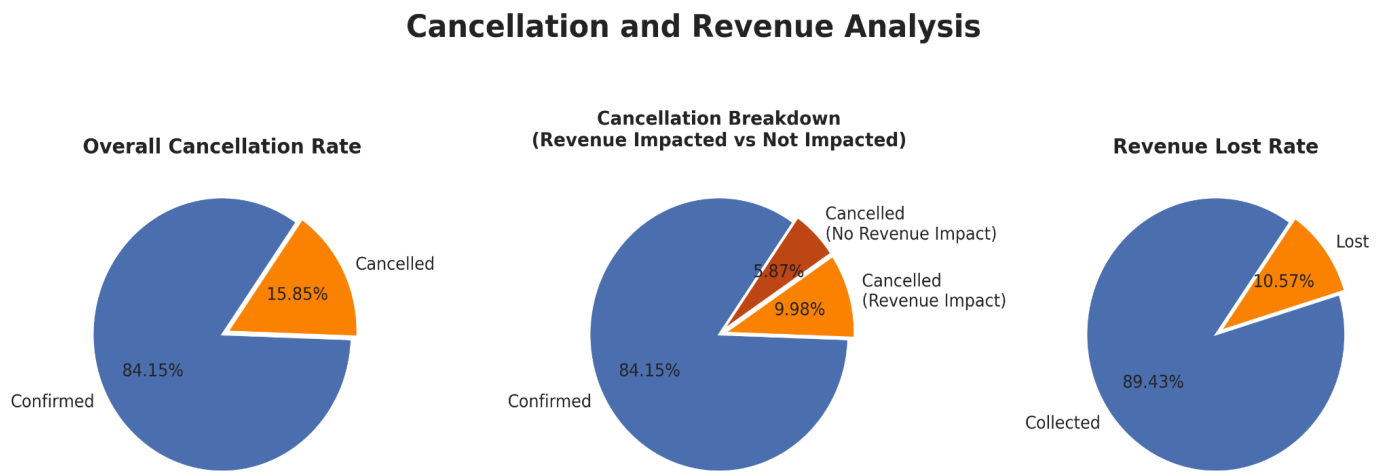
- **Data and Time Period:** Amazon India sales data for the full year 2022, containing details orders, status, amount, product category, SKU, size, fulfilment type, and shipping geography.

ANALYSIS SCOPE

- Dataset: 1,28,975 orders (Full year 2022)
- Period: January 1 - December 31, 2022
- Analysis Type: Descriptive (what happened)

KEY FINDINGS

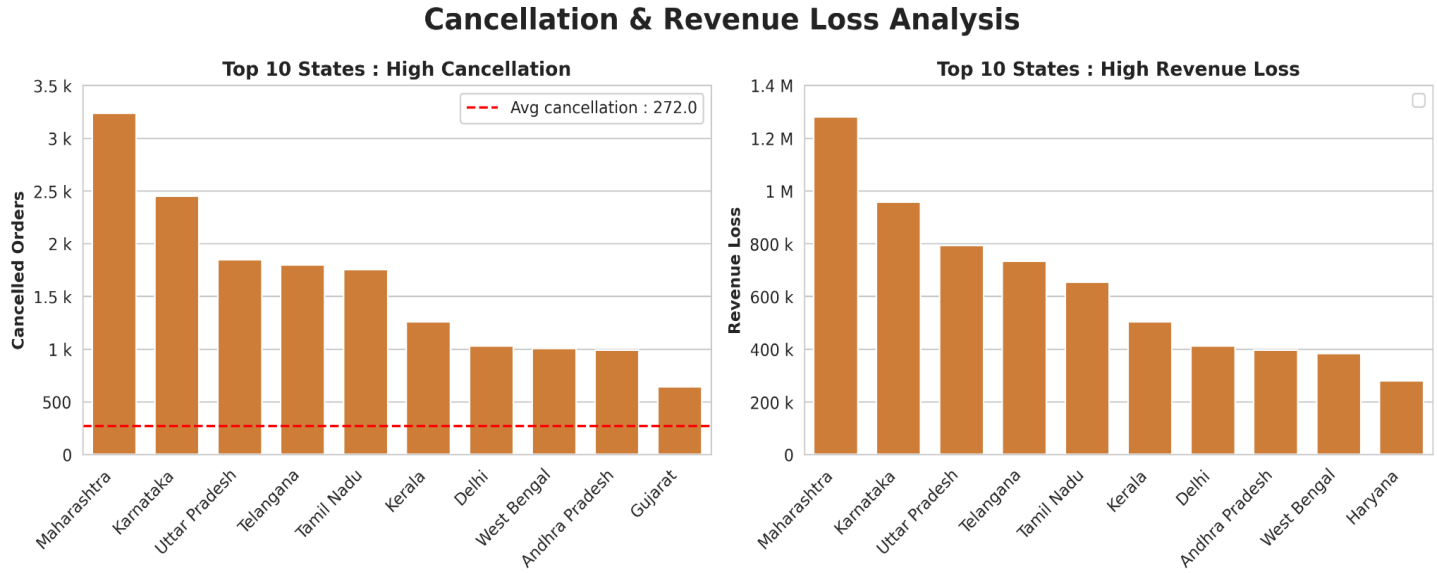
OVERALL CANCELLATION & REVENUE LOSS RATE



Overall Impact:

- Total Orders - 1,28,975
- Cancelled Orders - 20441 (12,872 orders with revenue impact + 7,569 without revenue impact)
- Cancellation Rate - 15.85% (~10% revenue impact + ~6 non revenue impact)
- Revenue Expected - ₹78592678.3
- Revenue Lost - ₹8,296,548.3
- Revenue Lost Rate - 10.57%
- Average revenue loss per cancelled order - ~₹406

GEOGRAPHIC ANALYSIS



Where is the problem?

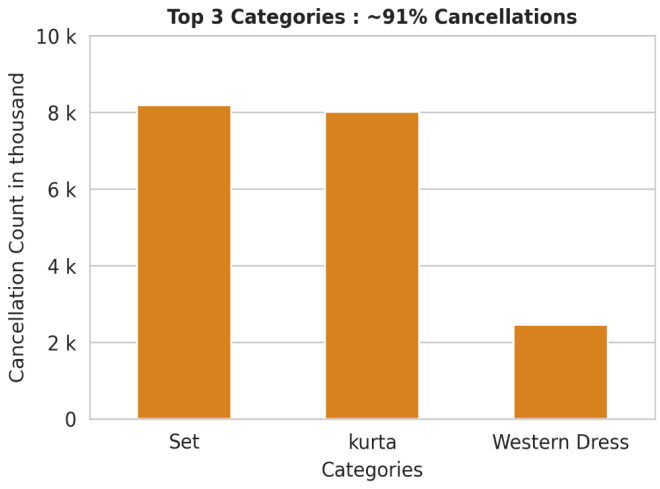
Observation - Cancellation is highly concentrated on high count not rate (Volume > Rate)

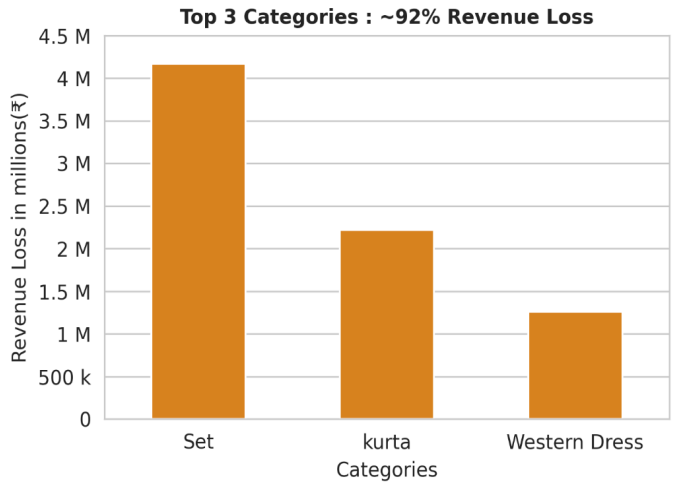
- Top 5 states (Maharashtra, Karnataka, Uttar Pradesh, Telangana, Tamil Nadu) drive ~54% of total cancellations and ~53% of total revenue loss though the cancellation rate and revenue loss rate of these states are lower than any other states.
- Top 5 highest contributing states have a cancellation rate between ~14-17% and revenue loss rate ~9-11%.
- Remote and Hills states, the lowest contributing states(Lakshadweep, Unknown, Dadar and Nagar, Ladakh, Mizoram) have the highest cancellation rate ~18-30% and revenue loss rate is ~19-23%.
- Highest Revenue loss state is **"Maharashtra"** wherein Lowest is **"Lakshdweep".**
- States have a cancellation rate between ~12-30% and Revenue loss rate between ~9-23%.
- Volume matters more than rates to business impact so focusing on Top 5 highest contributing states would save more than 50% of business problems.

No	Top 5 States	Cancellation count	Cancellation (%)	Contribution Cancellation(%)	Revenue Lost	Revenue Lost(%)	Contribution Loss(%)
1	Maharashtra	3236	14.54	15.83	1280597.14	9.6	15.42
2	Karnataka	2452	14.15	12.0	957303.37	9.13	11.52
3	Uttar Pradesh	1847	17.36	9.04	792999.08	11.63	9.54
4	Telangana	1800	15.89	8.81	732103.65	10.58	8.81
5	Tamil Nadu	1752	15.26	8.57	653868.11	10.04	7.87
	Total	**11087**		**54.25%**	**4416871.3**		**53.19%**

CATEGORICAL ANALYSIS

Cancellation & Revenue Loss Analysis





Which categories are impacting business more?

Observation -

- Top 3 problem categories (Set, Kurta and Western Dress) drive ~91% cancellation and ~92% revenue loss and Other 6 categories only contributes ~8-9% to whole cancellation and revenue loss, focusing on Top 3 categories would save almost the business impact

- Top 3 categories has ~15-16% cancellation rate and ~10-11% revenue loss rates

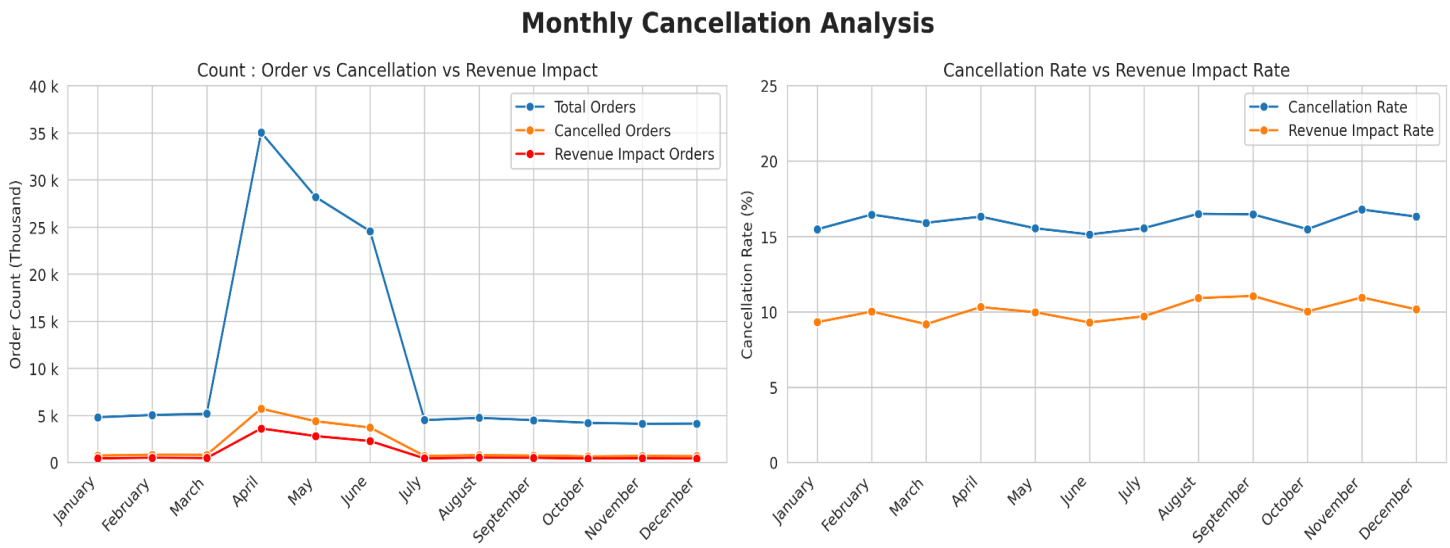
- Categories have a Cancellation rate between ~13-16% and Revenue Loss rate between ~8-11%.

- Highest Revenue loss category is **"Set"**, wherein Lowest is **"Saree"**

- Set alone contributes ~50% revenue loss, fixing Set category would solve 50% business impact.

No	Top 3 Category	Cancellation Count	Cancellation (%)	Contribution (%)	Revenue Loss	Revenue Loss(%)	Contribution Loss(%)
1	Set	8182	16.27	40.02	4171603.03	10.64	50.23
2	kurta	8007	16.05	39.17	2218279.7	10.41	26.71
3	Western Dress	2465	15.90	12.05	1257522.69	11.21	15.14
	Total	18654		**91.25%**	7647405.42		**92.09%**

TIME TREND ANALYSIS



Is the problem worse, improving or stable?

Observation -

- Cancellation rate is between ~15-16% and impact rate is between ~9-11% throughout the year there is no spike in cancellation rate but during April to June, there is a spike observed in absolute number.

- The cancellation rate is stable throughout the year and it's a strategic/process issue.

- No seasonal spikes in rates though during April - June ~7x spike in absolute numbers

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DATA CLEANING PERFORMED

- Found 69 unique state values (expected ~28-36 Indian states), cleaned variations (e.g., "rajeshthan" → "Rajasthan", "ap" → "Arunachal Pradesh", "pb" → "Punjab")

- Missing states are flagged as "Unknown", final states 37 including "Unknown"

- Missing amount is considered as cancelled if "status" contains (Cancelled, Returned and Rejected)

DEFINITIONS

Cancellation - Cancelled orders are considered where orders are **Cancelled**, **Returned**, **Damaged** or **Rejected**.

Revenue Lost - Cancelled orders with Amount > 0. (Excludes ₹0 and missing amounts)

LIMITATIONS

- ~6% amount is missing though almost all orders are cancelled but there are some valid orders
- Ship state has 0.03% missing
- No cancellation reason codes available
- Cannot differentiate customer vs. system-initiated cancellations